IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Prepared by: Le Phu
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Outline

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- Methodology
- Results
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Executive Summary

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Goal: predict Falcon 9 first-stage landing outcomes to support launch-cost decisions.
Data scope: 90 launches, 3 launch sites, 11 orbit classes.
Overall landing success rate: 66.7%.
Best model: DecisionTree with 88.9% test accuracy.
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- Objective: predict Falcon 9 first-stage landing outcomes for cost/risk planning.
- Dataset: 90 launches across 3 launch sites.
- Landing success rate: 66.7%.
- Best classifier: DecisionTree (88.9% accuracy).

Introduction

Business question: can first-stage landing success be predicted before launch?
Why it matters: landing probability is a key cost and reliability signal.
Deliverables: data collection, wrangling, EDA/SQL, interactive analysis, classification model.

- Problem: estimate landing success probability before launch.
- Context: reusable first-stage recovery drives launch economics.
- Scope: data collection, wrangling, EDA/SQL, interactive maps, and classification modeling.

Section 1

Methodology

Methodology

Method integrates API + scraping data, feature preparation, EDA/SQL validation, and ML testing.

- Each stage produces evidence used directly in the report slides.

API notebook: collect the data/jupyter-labs-spacex-data-collection-api.ipynb.
Scraping notebook: collect the data/jupyter-labs-webscraping.ipynb.
Primary files used: data/dataset_part_1.csv and data/spacex_web_scraped_1-Copy1.csv.

- Collected launch data from SpaceX API and Wikipedia-derived tables.
- Unified records into consistent fields for mission, payload, orbit, and landing outcome.

Called SpaceX REST endpoints and extracted launch, payload, pad, and core fields. Normalized nested JSON into tabular features for analysis. Output supports repeatable refresh when API data updates.

- API pipeline: launch endpoint retrieval, nested-field expansion, and table normalization.
- Output supports reproducible updates as source data evolves.

Scraped Falcon launch tables with requests + BeautifulSoup.
Parsed date, booster version, payload, orbit, and outcome columns.
Used as a secondary source to cross-check API records.

- Web-scraping pipeline: table extraction, parsing, and schema alignment.
- Used as cross-check source to validate key launch attributes.

Data Wrangling

- Wrangling steps: null handling, label engineering, and feature consistency checks.
- Prepared modeling-ready dataset with numeric and categorical inputs.

Visual EDA tested relationships among launch site, payload mass, orbit type, and flight number. Observed trend signals that support model feature relevance.

- Scatter plots tested relationships among flight number, payload, orbit, and launch site.
- Bar/line charts summarized orbit success and yearly landing trend.
- Notebook reference: eda/edadataviz.ipynb

Queried launch-site coverage, payload aggregates, mission outcomes, and date filters. Confirmed business-level metrics such as NASA payload totals and outcome counts. SQL outputs used as evidence slides for rubric alignment.

- SQL EDA answered launch-site coverage, payload aggregates, and mission-outcome questions.
- Query outputs provide auditable evidence for business conclusions.

Build an Interactive Map with Folium

- Folium map includes launch-site markers, clusters, and distance overlays.
- Interactive geography supports interpretation of site-level outcome behavior.

Build a Dashboard with Plotly Dash

- Dashboard design: filter by launch site and payload range for rapid comparison.
- Core outputs: success-count pies and payload-vs-outcome scatter view.

Predictive Analysis (Classification)

- Built and compared Logistic Regression, SVM, Decision Tree, and KNN.
- Used held-out test evaluation to select the final model.

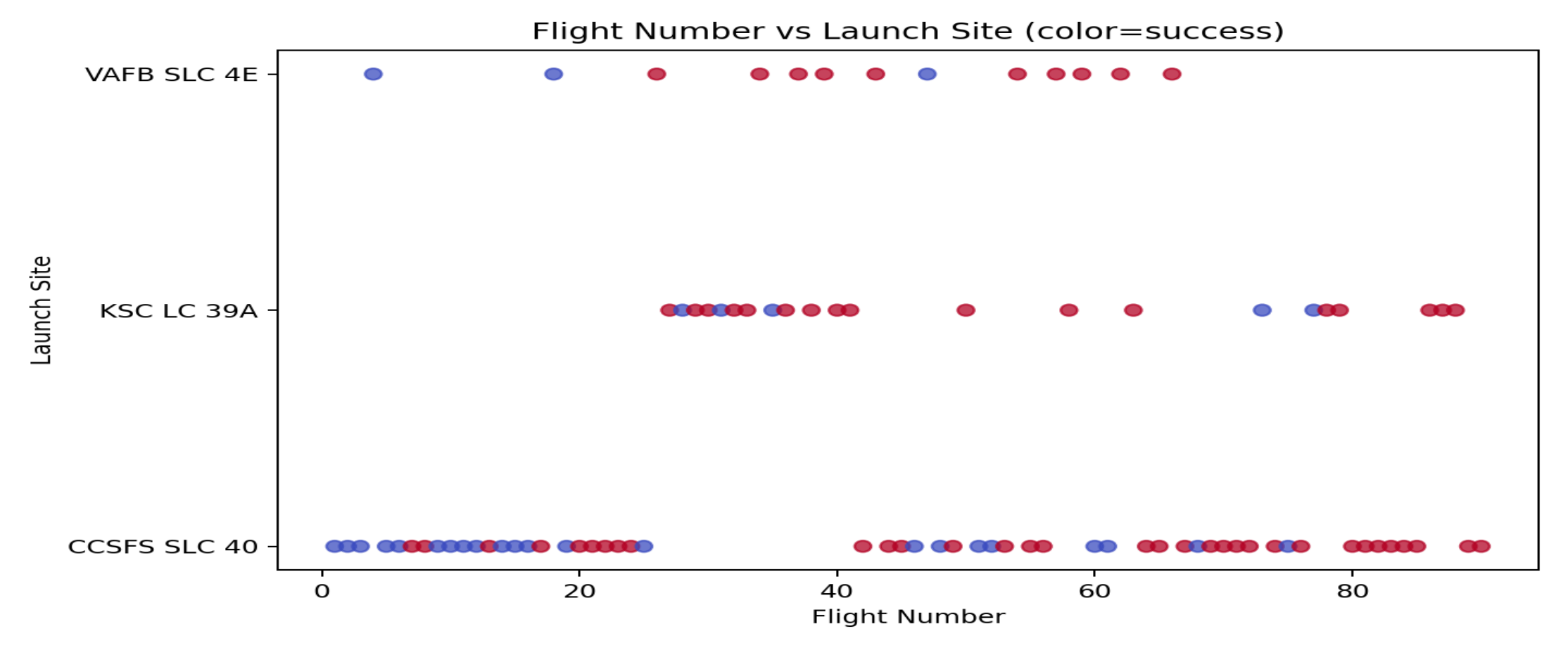
Results

- Results summarize EDA evidence, interactive analytics, and predictive performance.
- Conclusions are tied to observed trends and model validation metrics.

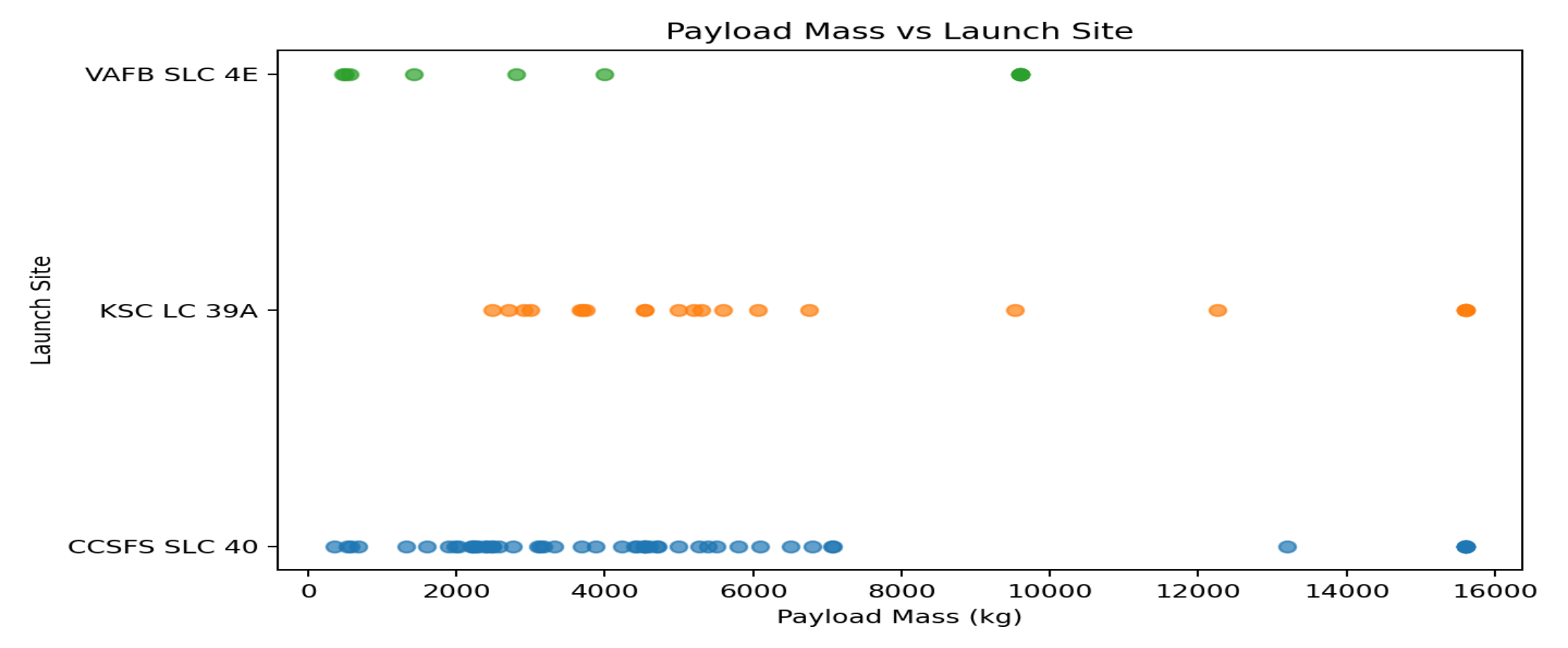
Section 2

Insights drawn from EDA

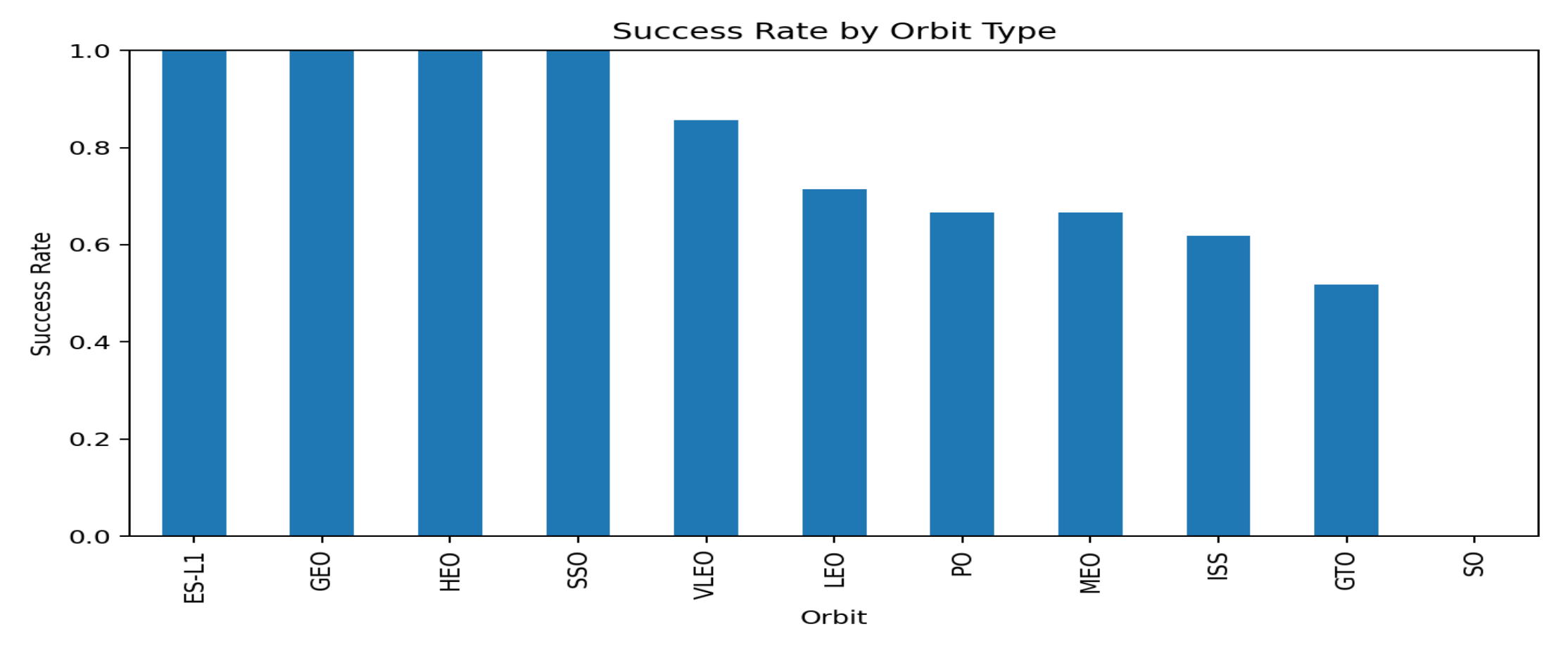
Flight Number vs. Launch Site



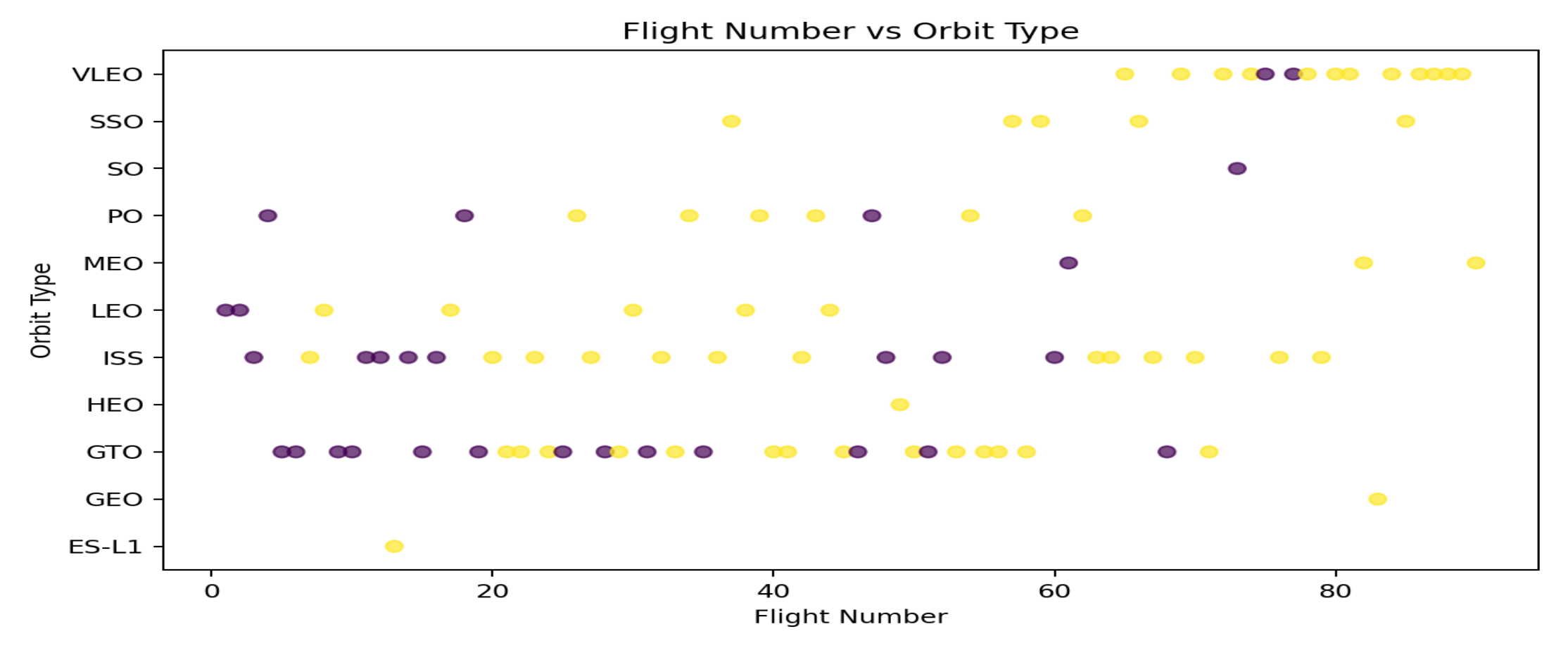
Payload vs. Launch Site



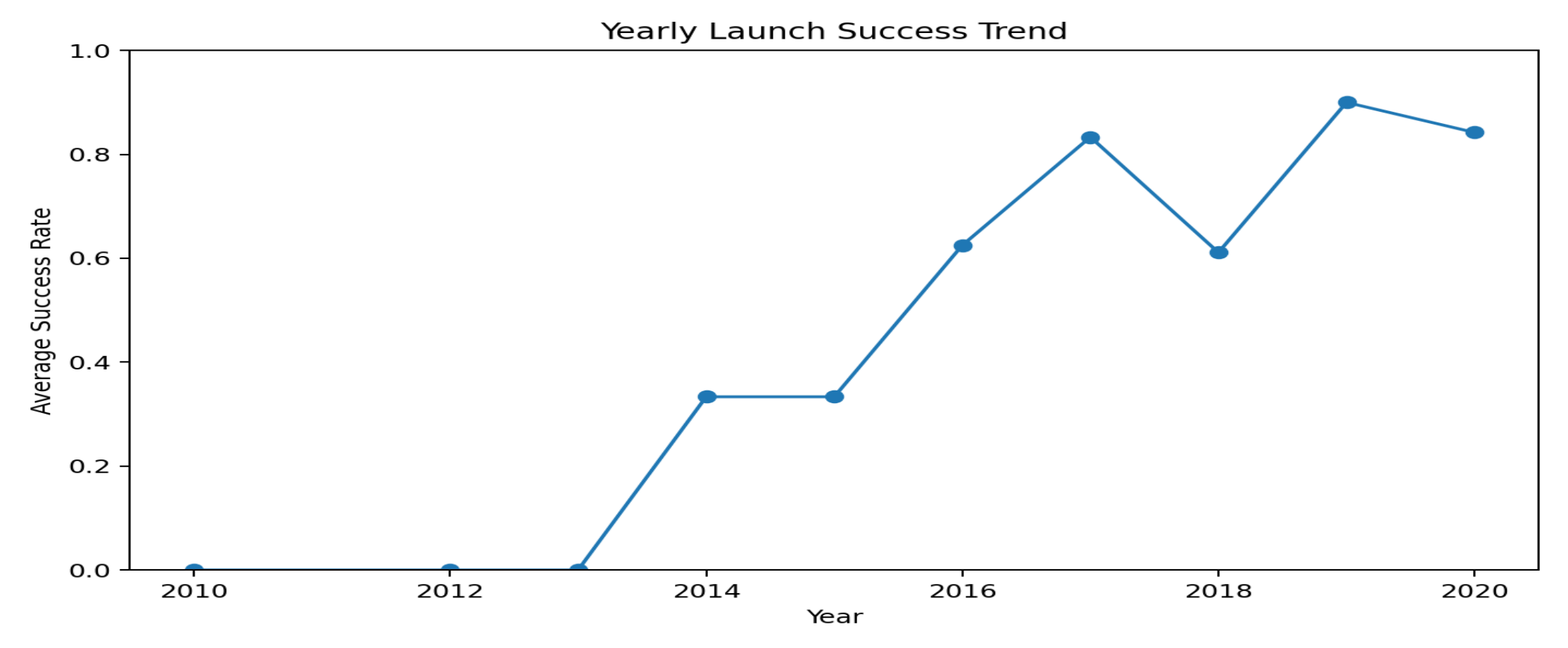
Success Rate vs. Orbit Type



Flight Number vs. Orbit Type



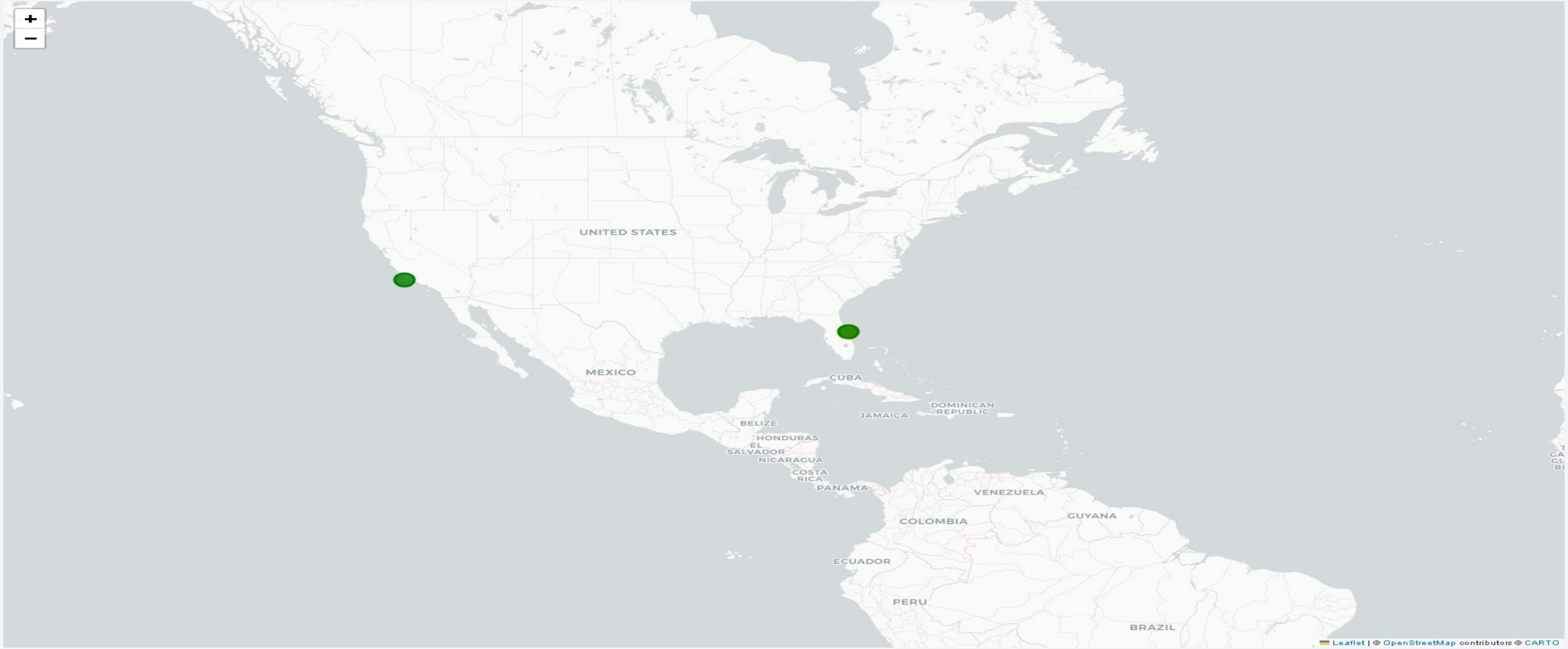
Launch Success Yearly Trend



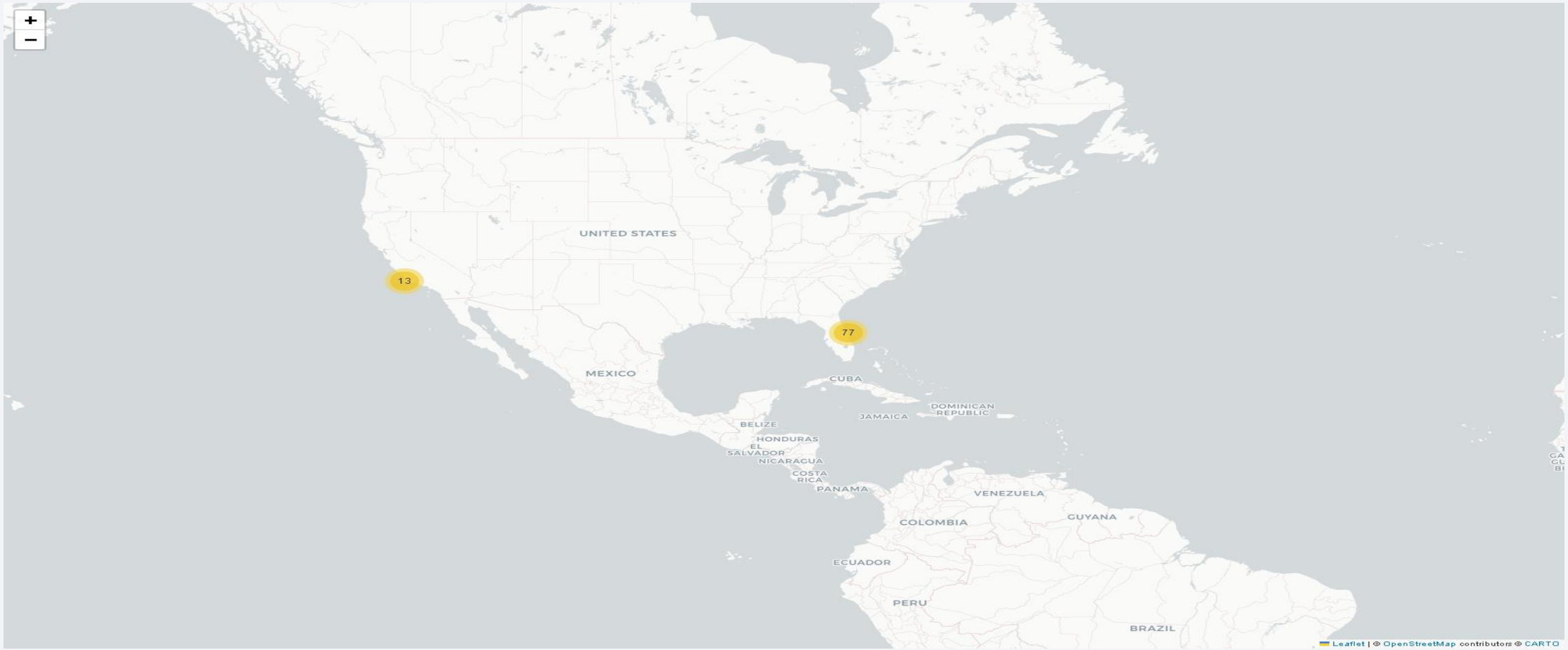
Section 3

Launch Sites Proximities Analysis

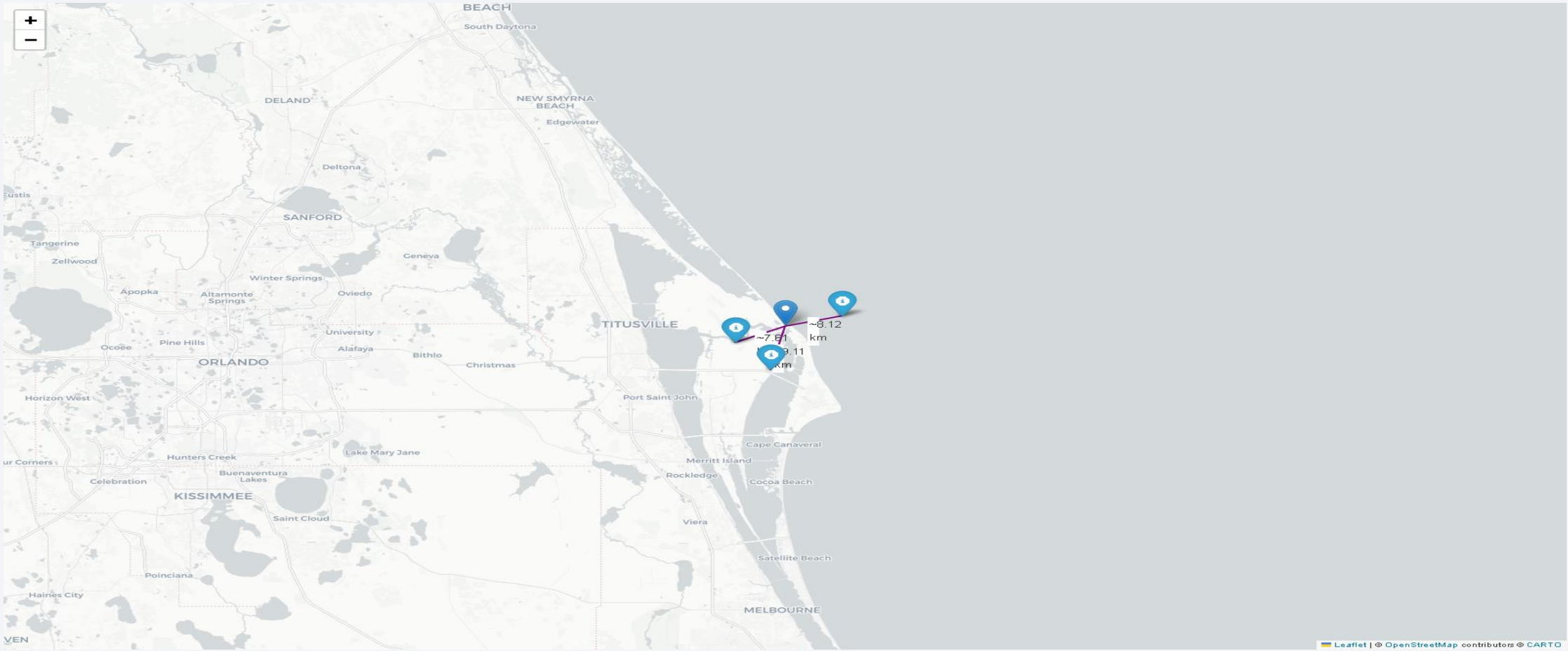
Launch Sites and Success Rate Markers



Clustered Launch Outcomes



Distance Overlays from Reference Site



Section 4

Build a Dashboard with Plotly Dash

Section 5

Predictive Analysis (Classification)

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Model comparison (test accuracy): LogReg=0.778, SVM=0.778, DecisionTree=0.889, KNN=0.833.
Selected best model: DecisionTree.
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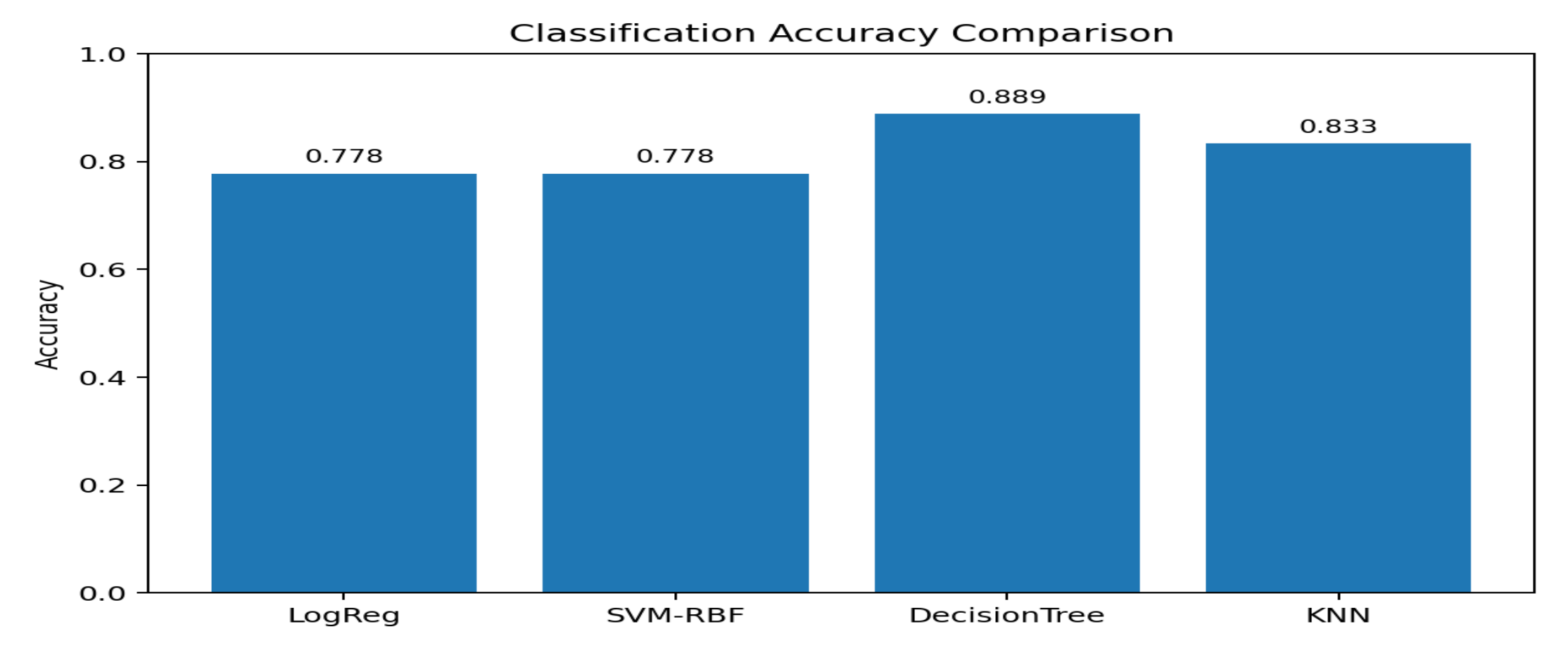
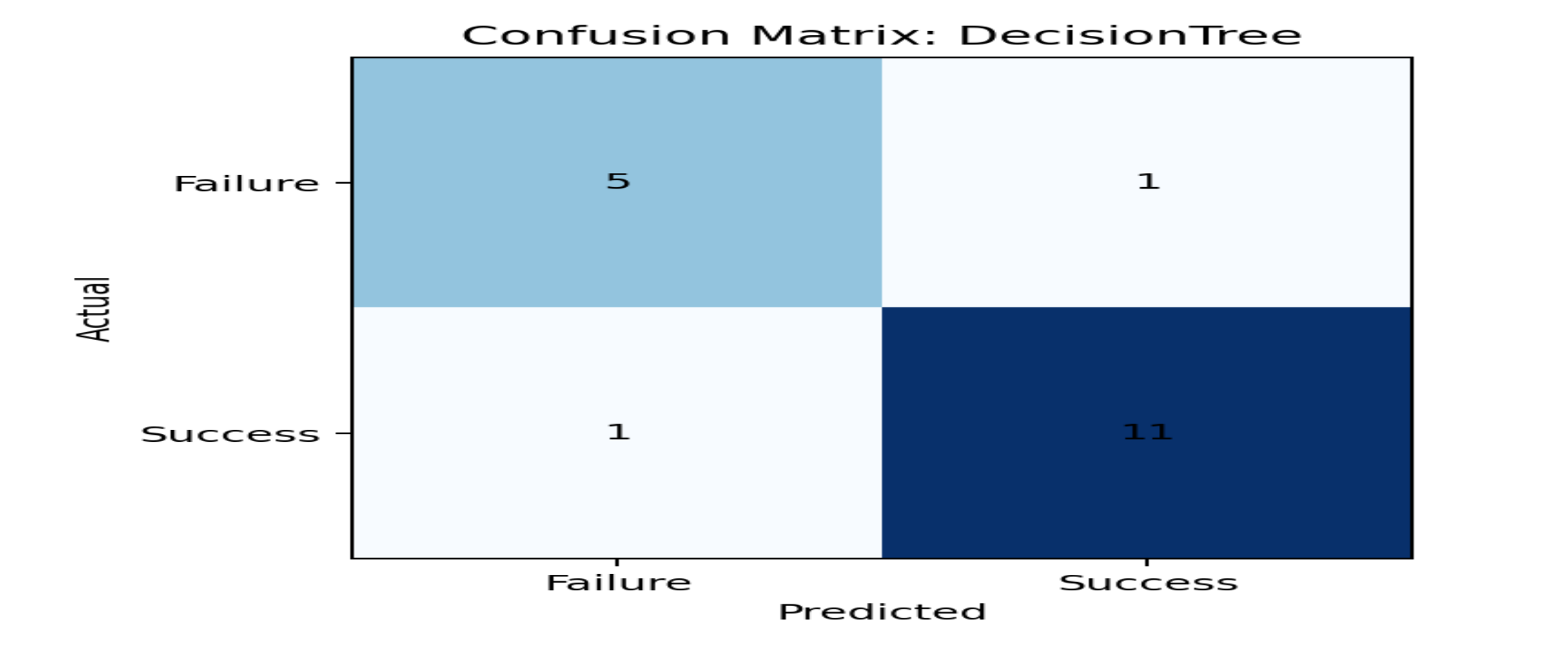


Confusion Matrix



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Landing success improved substantially in recent years.
Orbit type, launch site, and mission progression are informative predictors.
Decision Tree achieved the best held-out performance (88.9%).
Recommendation: keep model retraining pipeline and add richer weather/telemetry features.
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- Landing success increased over time and varies by orbit/site context.
- Best test model: DecisionTree with 88.9% accuracy.
- Next step: enrich features (weather/telemetry) and retrain for stronger generalization.

Appendix

- Assets included in this report: report/assets/*.png, report/assets/analysis_summary.json, notebooks under collect the data/, eda/, and interactive visual/.

Thank you!